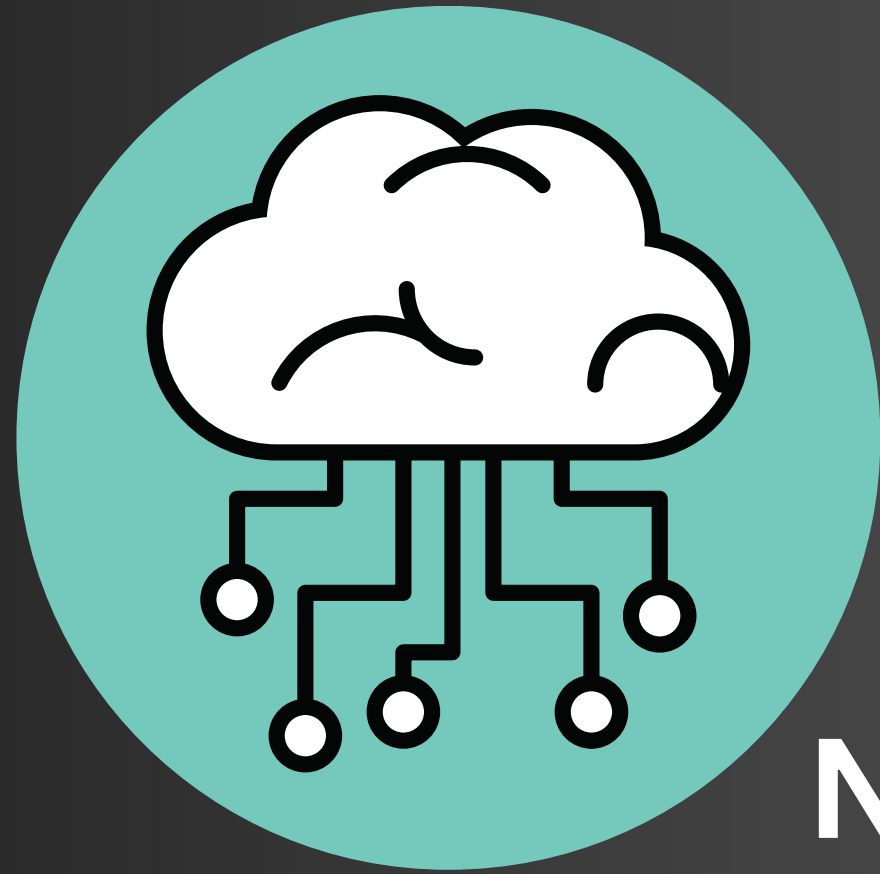
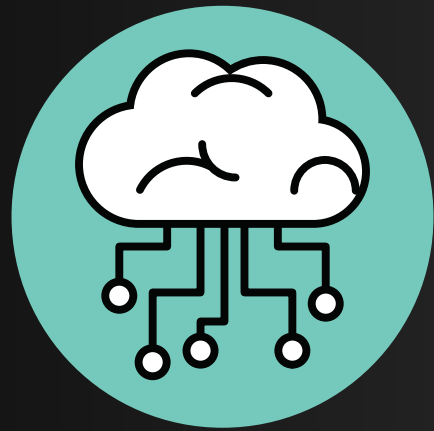




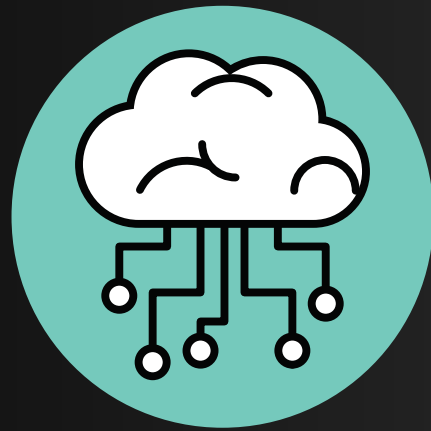
Networking

Virtual Private Cloud (VPC)



Virtual Private Cloud (VPC)

A private, isolated network within the AWS cloud where you can launch and manage your resources securely.



Why we need VPC?

To securely isolate and control network environments.





**Website is ready
Where to deploy?**



**Website is ready
Where to deploy?**





**Website is ready
Where to deploy?**



US

Asia

Europe



**Website is ready
Where to deploy?**



Asia

North

East

South

REGION



Website is ready
Where to deploy?



Asia

North

Singapore

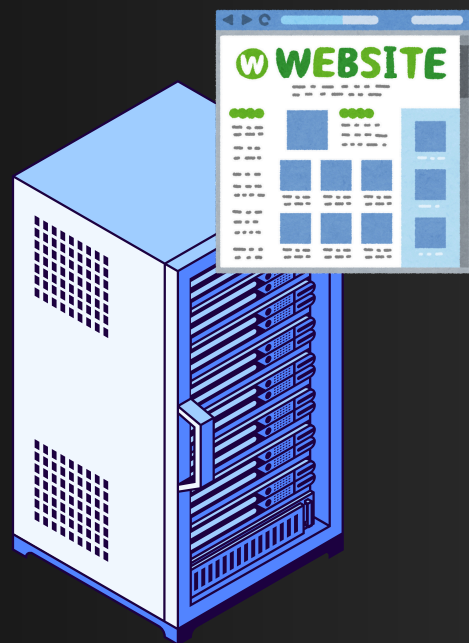
East

Mumbai
Hyderabad

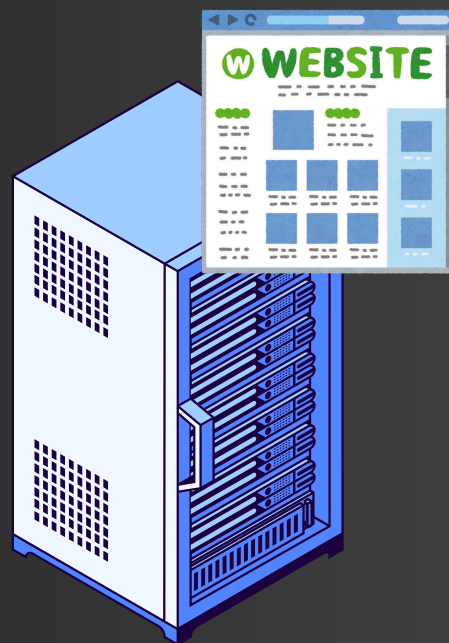
South

Tokyo

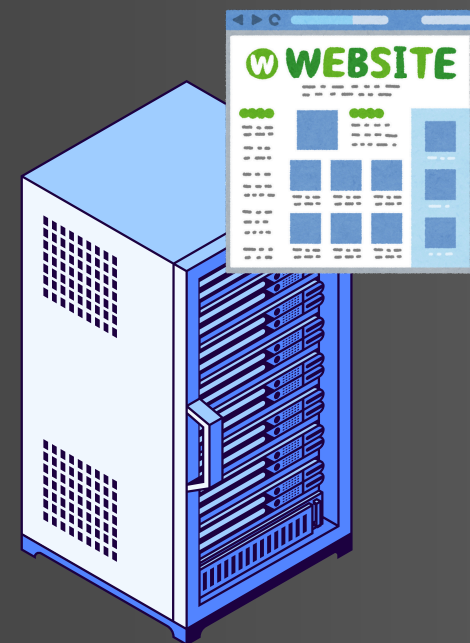
Availability Zones



a



b

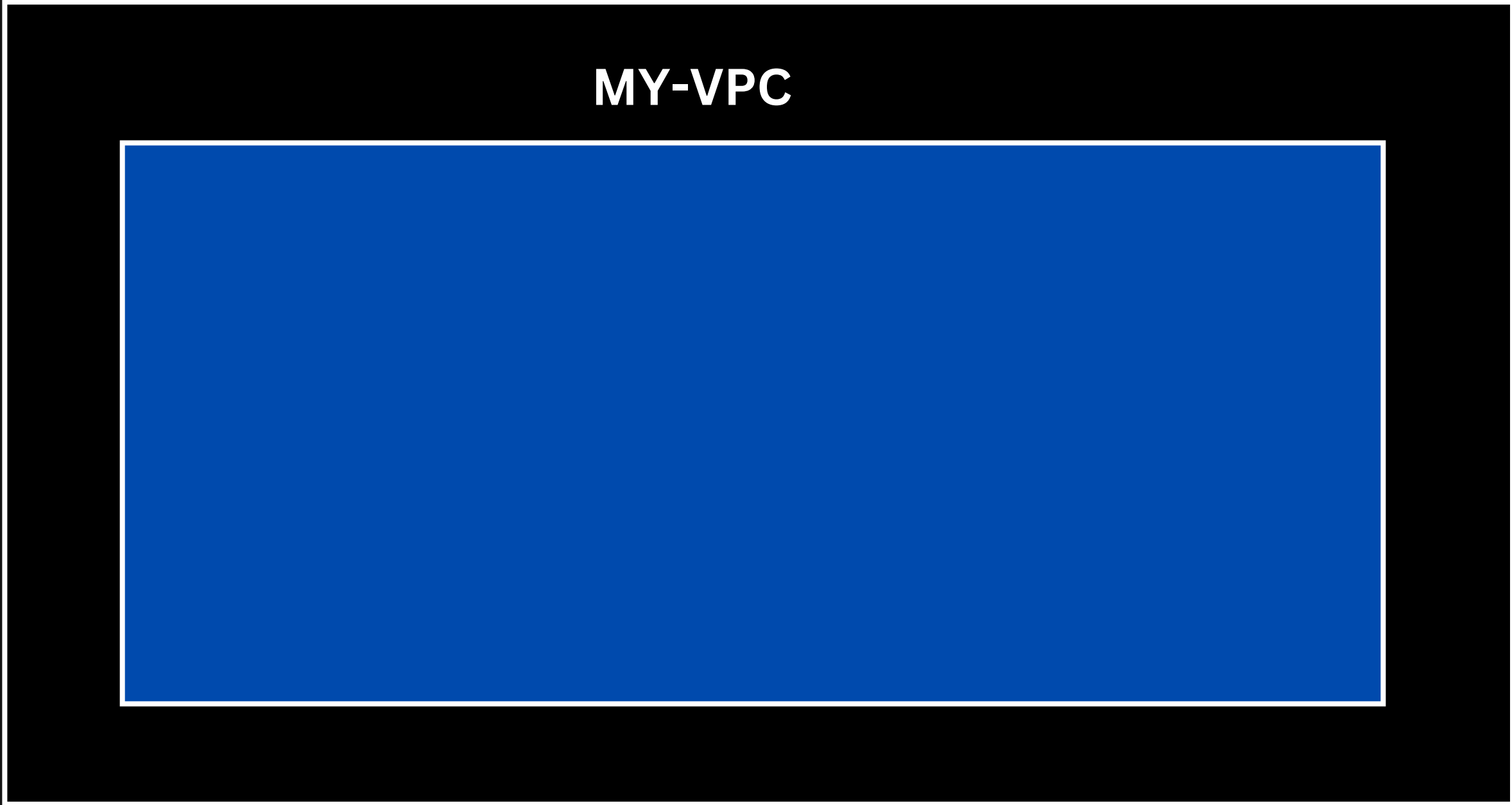


c

Mumbai

region - mumbai

MY-VPC

A diagram illustrating a nested structure. It consists of a large black rectangle with a white border. Inside this black rectangle, there is a smaller blue rectangle, also with a white border. The text "MY-VPC" is centered at the top of the black rectangle, above the blue rectangle.

VPC CIDR Block

When you create a VPC, you specify a CIDR block that defines the IP address range for the entire VPC. For example:

```
sh
```

[!\[\]\(4729e517bc6a7cd81c8025b9646574fb_img.jpg\) Copy code](#)

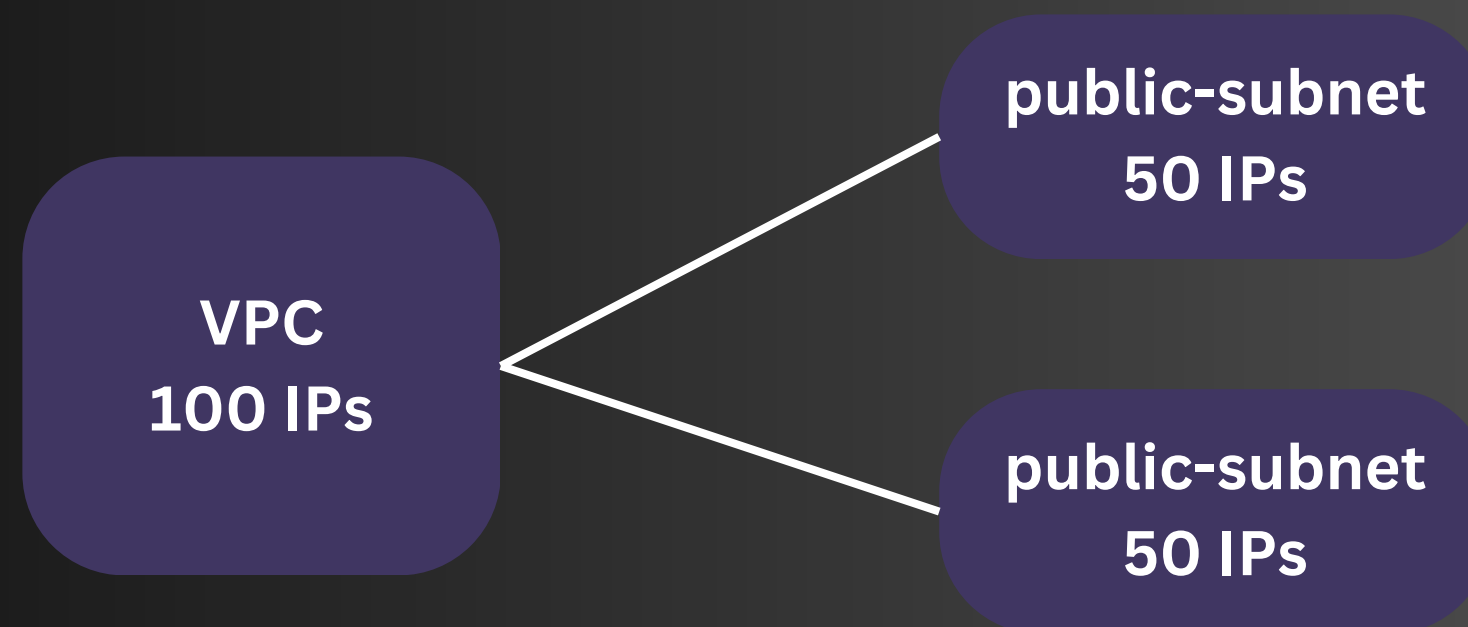
```
10.0.0.0/16
```

This block allows for 65,536 IP addresses (but in reality, 65,531 usable addresses).

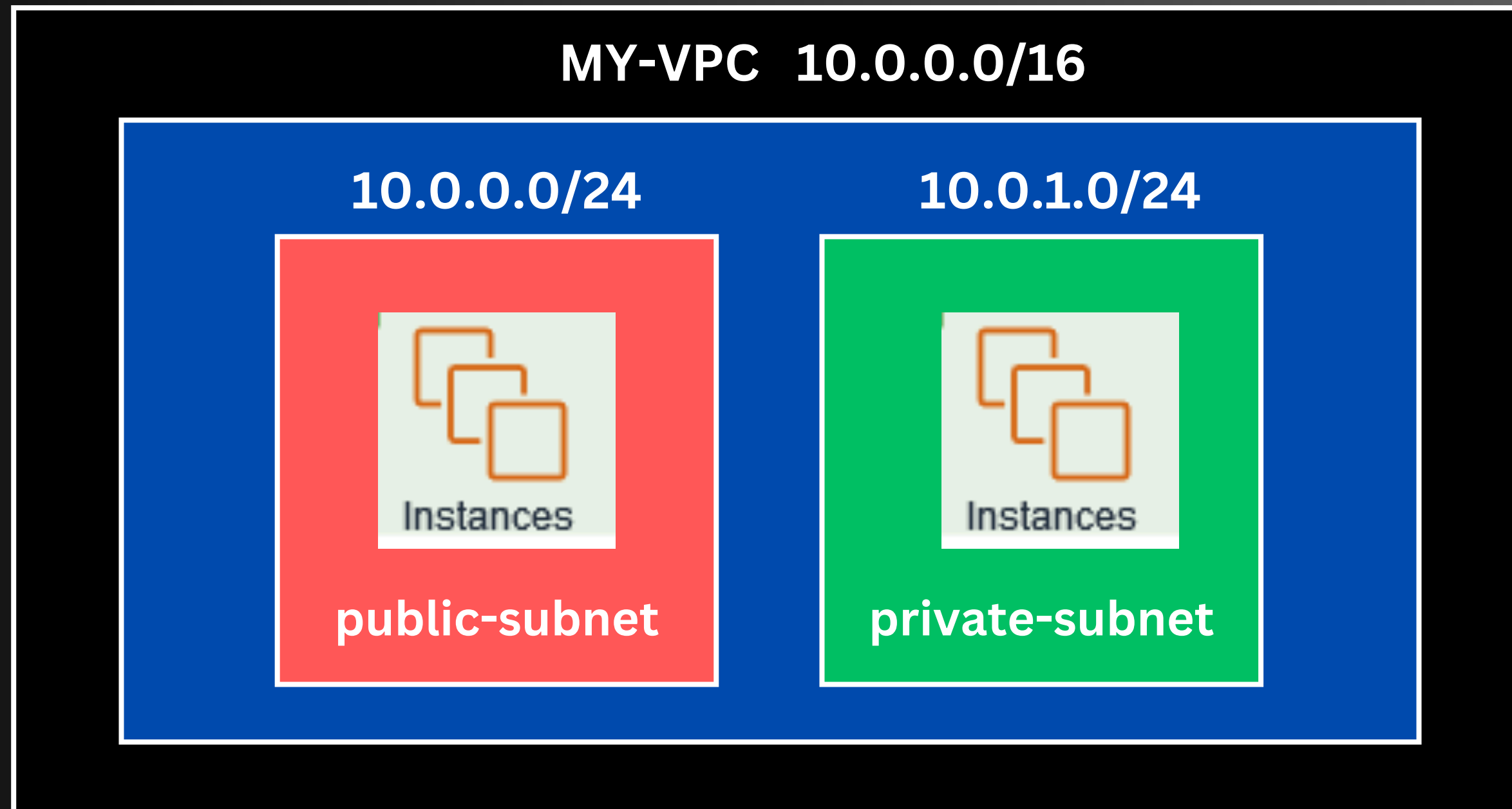
CIDR (Classless Inter-Domain Routing) is a method for allocating IP addresses and routing Internet Protocol (IP) packets.

What is Subnets?

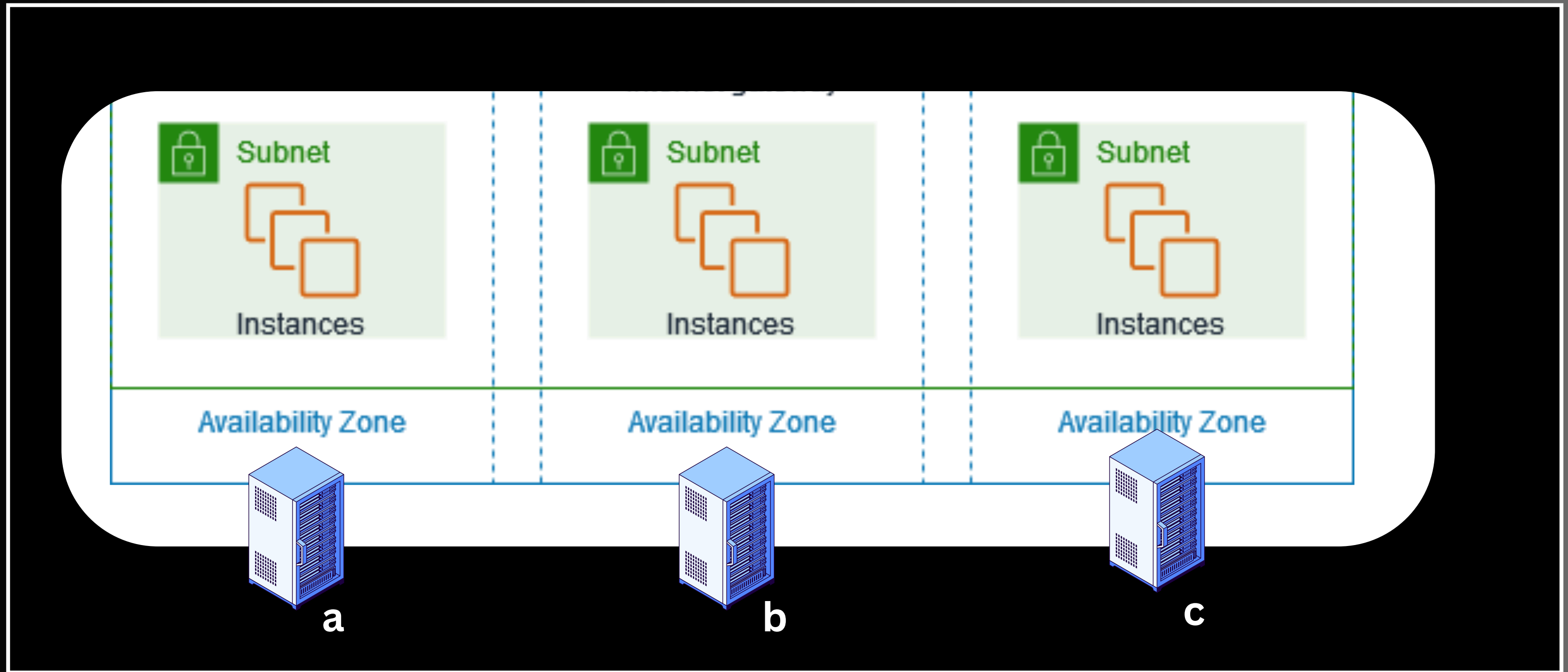
A subnet is a smaller, segmented part of a larger network that isolates and organizes devices within a specific IP address range.



region



region



What happens when creating subnet?

CIDR Block Allocation:

You specify a range of IP addresses (CIDR block) within the VPC's IP address range for the subnet.

This determines the pool of IP addresses available for instances in the subnet.

Subnet CIDR Blocks

Within the VPC, you can create subnets by allocating smaller CIDR blocks from the VPC's range.

For example:

- Public Subnet: ``10.0.1.0/24``
- Private Subnet: ``10.0.2.0/24``

Each of these subnets has 256 IP addresses (251 usable).

Explanation of 10.0.1.0/24

- An IPv4 address is 32 bits long.
- Example in binary: ``10.0.1.0` -> `00001010.00000000.00000001.00000000``

- The /24 indicates that the first 24 bits are the network portion of the address.
- The remaining 8 bits are available for host addresses within the network.

10.0.1.0 to 10.0.1.255 is the full range.

Route Table

What is a Route Table?

A route table is a set of rules, called routes, that are used to determine where network traffic from your subnets or gateway is directed. Each subnet in your VPC must be associated with a route table, which controls the routing for that subnet.

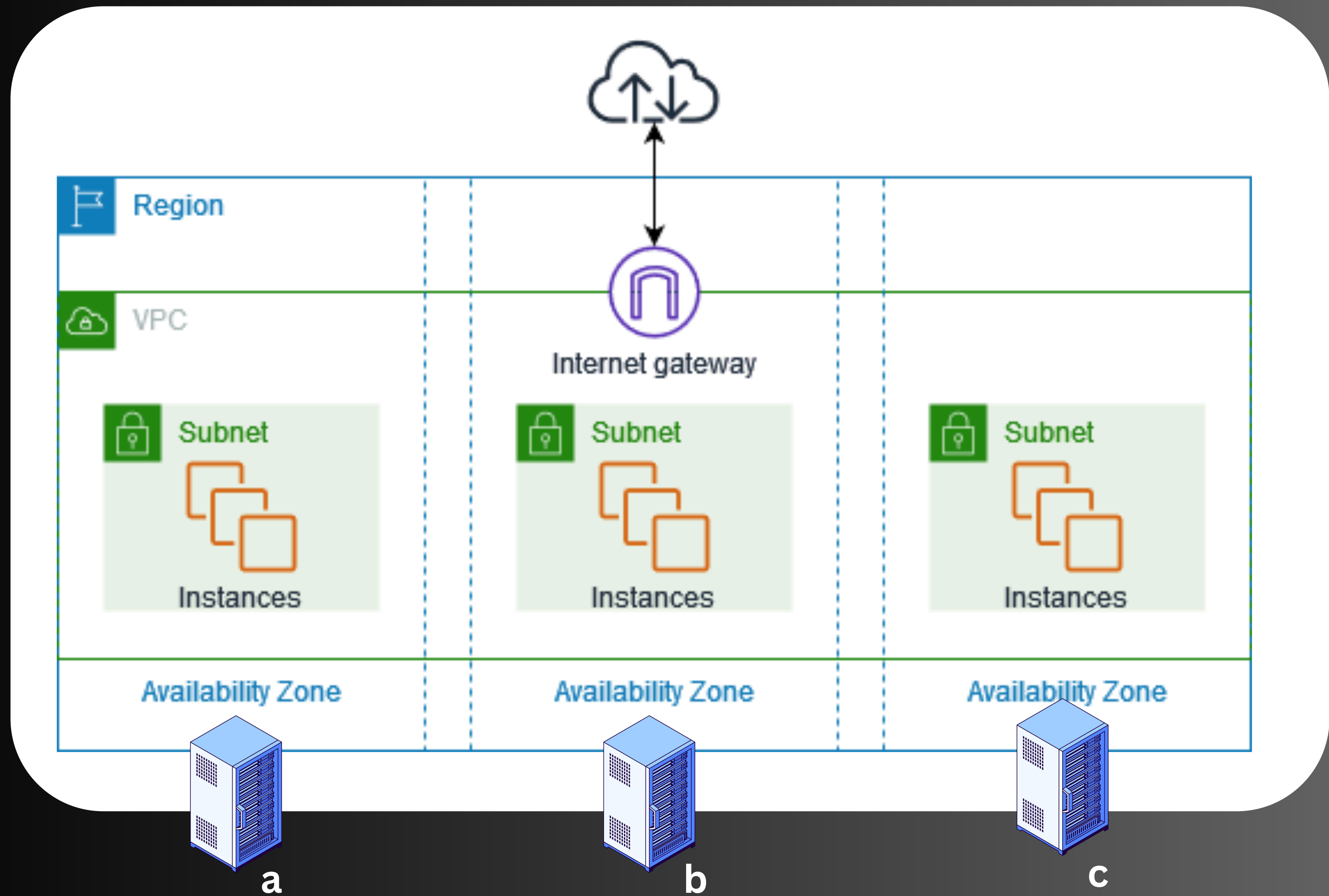
RouteTableDetails

Routes (2)	
Filter routes	
Destination	Target
0.0.0.0/0	igw-0bc1bb62e4e4f3c3c
172.31.0.0/16	local

Internet Gateway

An Internet Gateway is a component that allows communication between instances in your VPC and the internet.





Security Groups: Network firewall rules that control inbound and outbound traffic for instances.



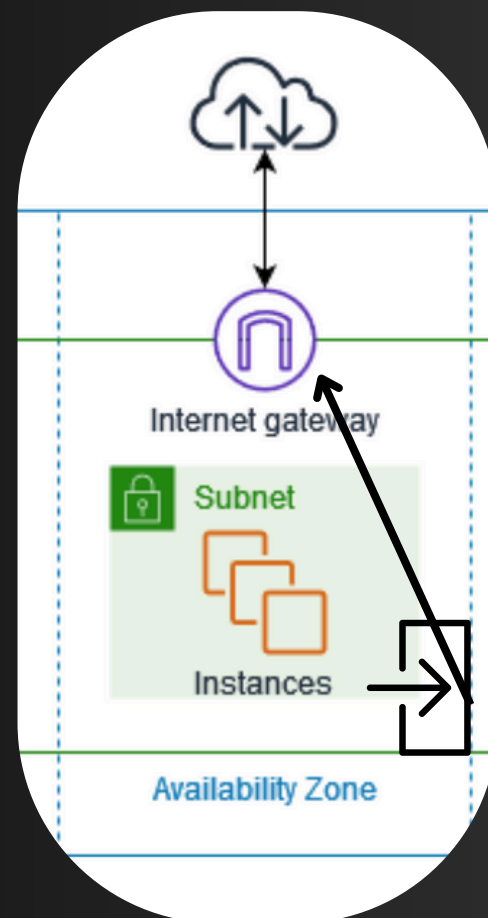
Network ACLs (Access Control Lists):

Optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets.

Allow or Deny Rule.

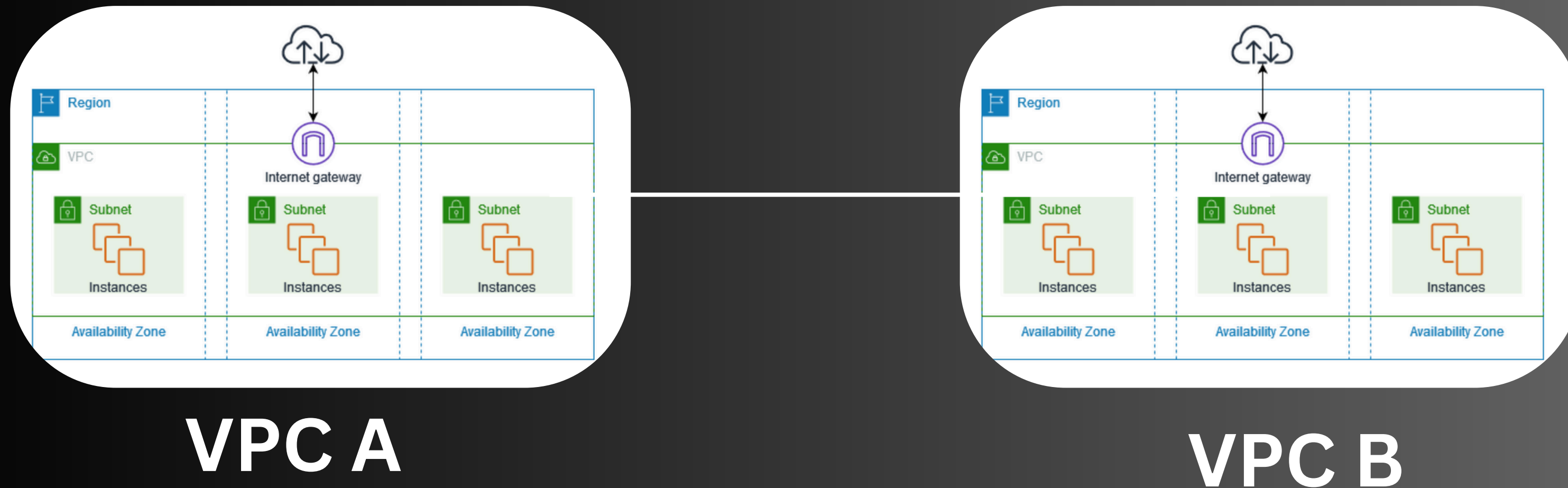
NAT (Network Address Translation) Gateway:

Enables instances in a private subnet to connect to the internet or other AWS services, but prevents the internet from initiating connections to those instances.

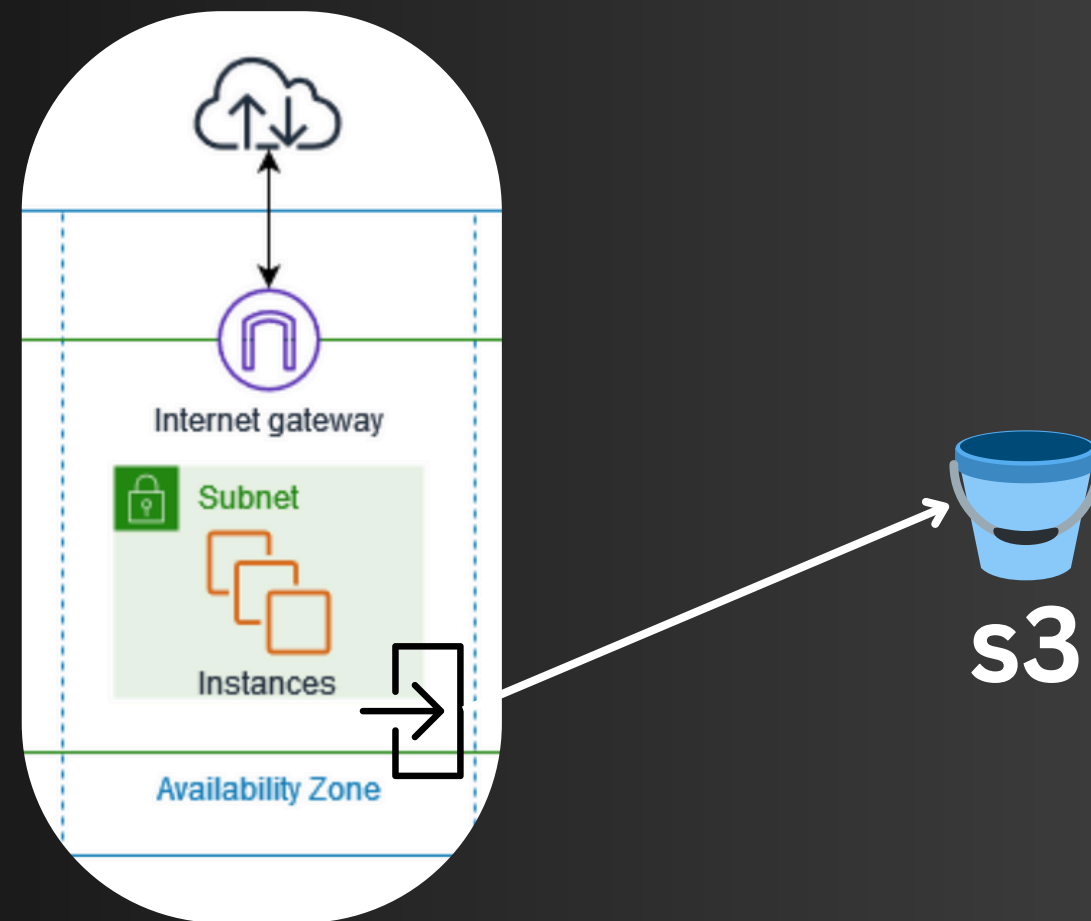


Downloading Updates

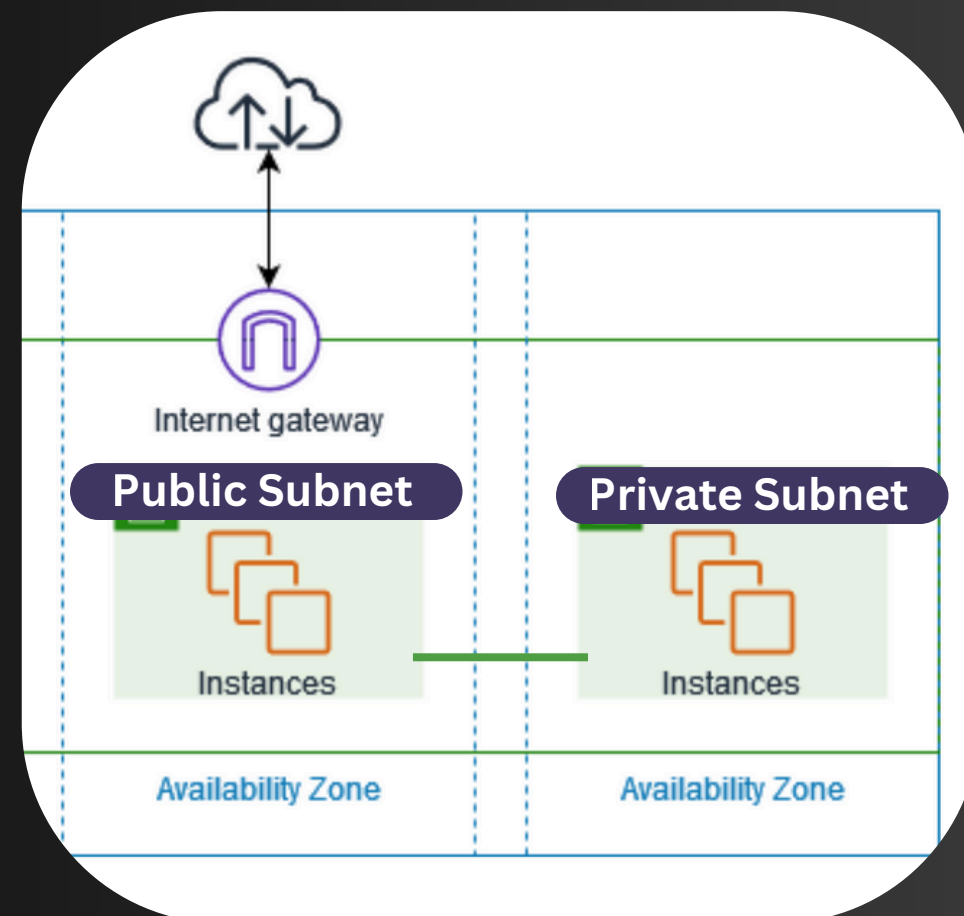
VPC Peering: A networking connection between two VPCs that enables you to route traffic between them privately.



VPC Endpoints: Allows you to privately connect your VPC to supported AWS services and VPC endpoint services powered by AWS PrivateLink.



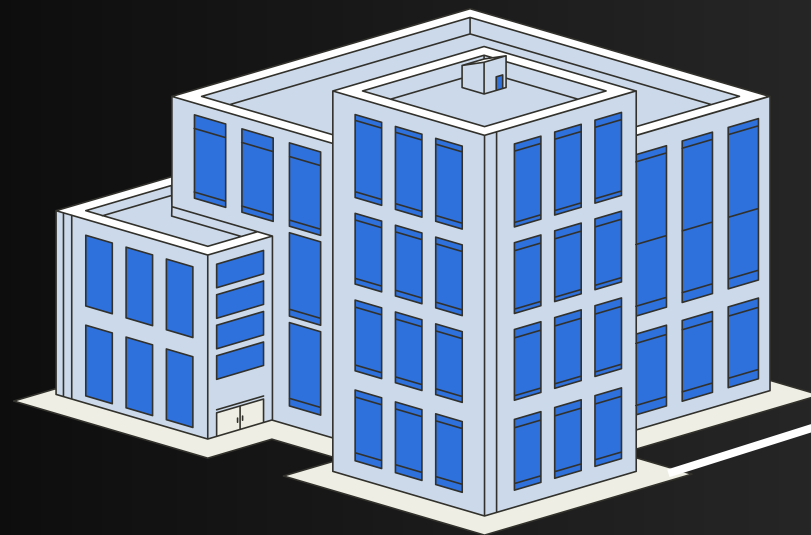
Bastion Host: A special-purpose instance that provides secure access to your instances in private subnets.



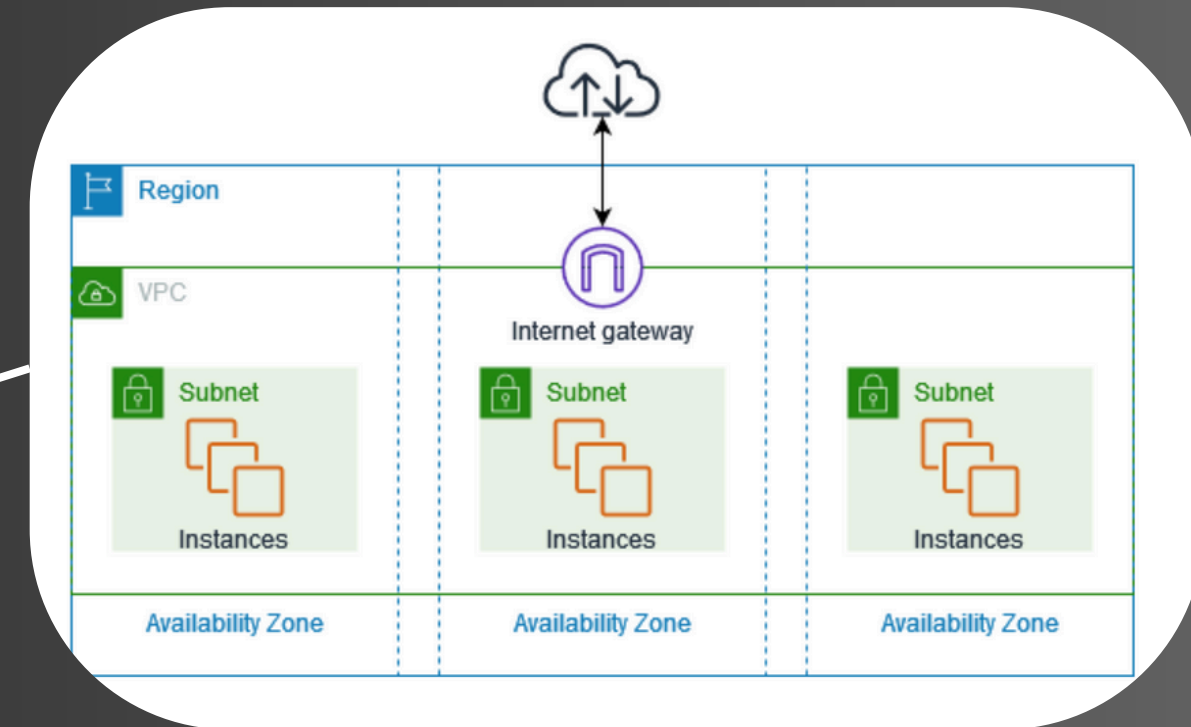
Elastic IP Addresses: Static IP addresses designed for dynamic cloud computing.

VPC Flow Logs: Capture information about the IP traffic going to and from network interfaces in your VPC.

Direct Connect: Establishes a dedicated network connection from your premises to AWS.



Office



Transit Gateway: A network transit hub that you can use to interconnect your VPCs and on-premises networks.

AWS Client VPN: Managed VPN service that enables secure remote access to AWS resources and on-premises networks using OpenVPN-based clients.

